# Smart Agriculture and Climate Resilience

Future Data Lab, Project 3

## 1. Problem Statement

Agriculture matters for food security and for the economy, but how well a crop does depends heavily on the weather and the soil. Rainfall, temperature, humidity and soil moisture all affect growth and yield, and they are becoming harder to predict. Rains come late, dry spells run longer, and temperatures swing more than they used to. Farmers and agricultural officers often have to decide what to plant, and where, before they know how the season will turn out. Too often those decisions rest on habit or guesswork instead of what past records show.

This project uses Python to investigate agricultural and environmental data and to find patterns in crop yields, rainfall and growing conditions across regions, crops and growing periods. We are not building a prediction model. The goal is to describe the patterns clearly enough that the findings can support better farming decisions and climate resilience planning.

## 2. Overall Objective

To investigate how environmental conditions (rainfall, temperature, humidity and soil moisture) relate to crop yield across regions, crops and growing periods, and to turn the findings into practical recommendations for climate-resilient farming.

## 3. Specific Objectives

1. To clean and validate the dataset (missing values, duplicates, impossible readings) so that the yield and weather figures can be trusted.
2. To compare crop yields across regions and crop types, and to analyse how rainfall and temperature vary across regions and growing periods.
3. To examine the relationship between environmental factors (rainfall, temperature, humidity and soil moisture) and crop yield, and to identify the conditions linked to high and low yields.

## 4. Research Questions

1. Which regions record the highest and lowest average crop yields?
2. Which crop types record the highest and lowest yields per hectare?

3. How does rainfall differ across regions, and across months or growing periods?
4. How do temperature and rainfall change across growing periods?
5. Is higher rainfall linked to higher yield, or does yield stop improving beyond a certain level?
6. How do temperature and humidity differ between high-yield and low-yield records?
7. Is soil moisture related to yield, and is the relationship the same for every crop?

## 5. Target Users

**1.Smallholder farmers:** to understand which conditions are linked to better and worse harvests and plan planting accordingly.

**2.Agricultural extension officers:** to give advice on crops and climate resilience that is backed by data.

**3.Agricultural policymakers and county planners:** to decide where to direct support during dry or unpredictable seasons.

**4.Agricultural researchers:** to spot patterns in the data that are worth studying further.